

Synoptic LE: Tree Fluxes 2022–2023

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1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with "remove" in the qaqc_flag columns
# stem_flux_2022_2023 <- stem_flux_2022_2023 %>%
#   filter(!grepl("remove", ch4_qaqc_flag, ignore.case = TRUE)) %>%
#   filter(!grepl("remove", co2_qaqc_flag, ignore.case = TRUE))

##Removed fluxes in the CH4 and CO2 flux processing Rmd code, removed both CH4 and CO2 obs w
```

```

r2_summary <- stem_flux_2022_2023 %>%
  summarise(total_obs = n(),
            n_neg_CO2 = sum(CO2_neg_flux , na.rm = TRUE),
            pct_neg_CO2 = 100 * n_neg_CO2 / total_obs,

            n_CO2_flux_rejected = sum(CO2_flux_rejected),
            pct_CO2_flux_rejected = 100 * n_CO2_flux_rejected / total_obs,

            n_CH4_rejected_model = sum(CH4_rejected_model),
            pct_CH4_rejected_model = 100 * n_CH4_rejected_model / total_obs)

#table
r2_summary %>%
  tidyr::pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(
    Metric = recode(Metric,
      total_obs          = "Total observations",
      n_neg_CO2          = "Negative CO2 flux (n)",
      pct_neg_CO2        = "Negative CO2 flux (%)",
      n_CO2_flux_rejected = "Rejected CO2 flux (n)",
      pct_CO2_flux_rejected = "Rejected CO2 flux (%)",
      n_CH4_rejected_model = "Rejected CH4 - poor model fit (n)",
      pct_CH4_rejected_model = "Rejected CH4 - poor model fit (%)"
    ),
    Value = ifelse(grepl("%", Metric),
      round(Value, 1),
      as.integer(Value))
  ) %>%
  knitr::kable(
    caption = "QA/QC CO2 and CH4 Flux Summary",
    align   = c("l", "r"),
    booktabs = TRUE,
    format   = "latex"
  ) %>%
  kableExtra::kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width     = FALSE
  ) %>%
  kableExtra::pack_rows("CO2", 2, 5) %>%
  kableExtra::pack_rows("CH4", 6, 7)

```

Table 1: QA/QC CO2 and CH4 Flux Summary

Metric	Value
Total observations	424.0
CO2	
Negative CO2 flux (n)	10.0
Negative CO2 flux (%)	2.4
Rejected CO2 flux (n)	15.0
Rejected CO2 flux (%)	3.5
CH4	
Rejected CH4 — poor model fit (n)	NA
Rejected CH4 — poor model fit (%)	NA

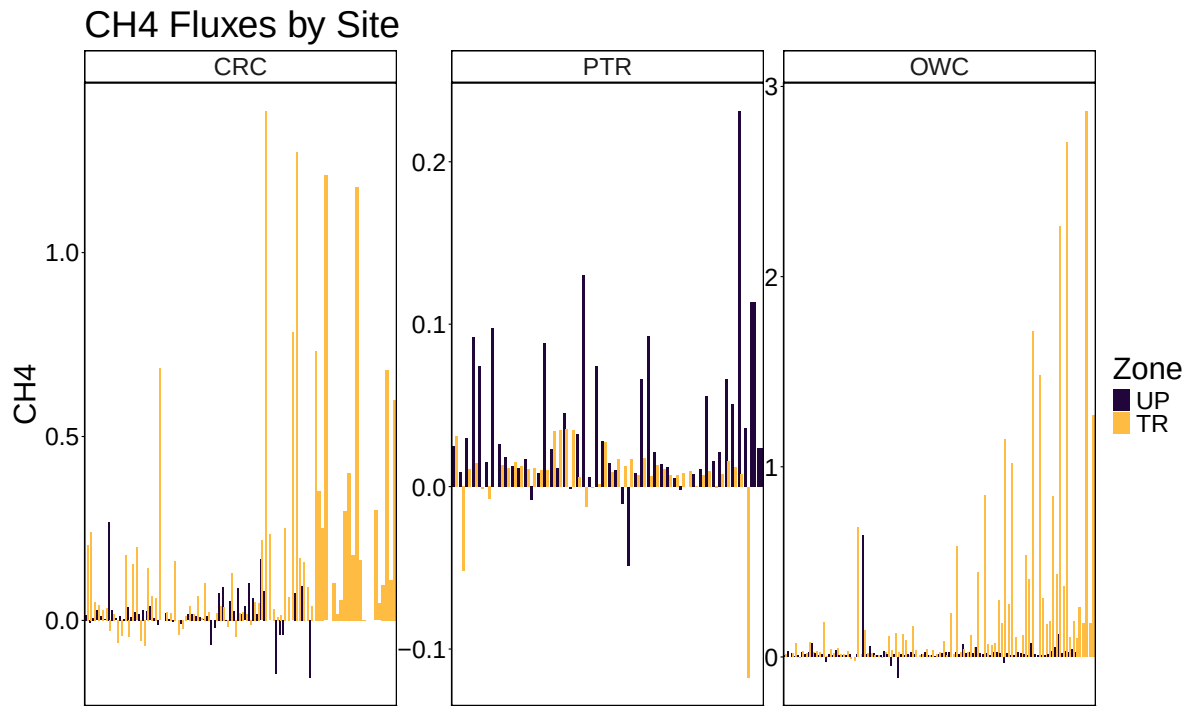
```
#removing negative CO2 fluxes from the final dataset but leaving flags for users to decide
stem_flux_2022_2023_filtered <-stem_flux_2022_2023%>%
  filter(co2_lin_flux.estimate>0)

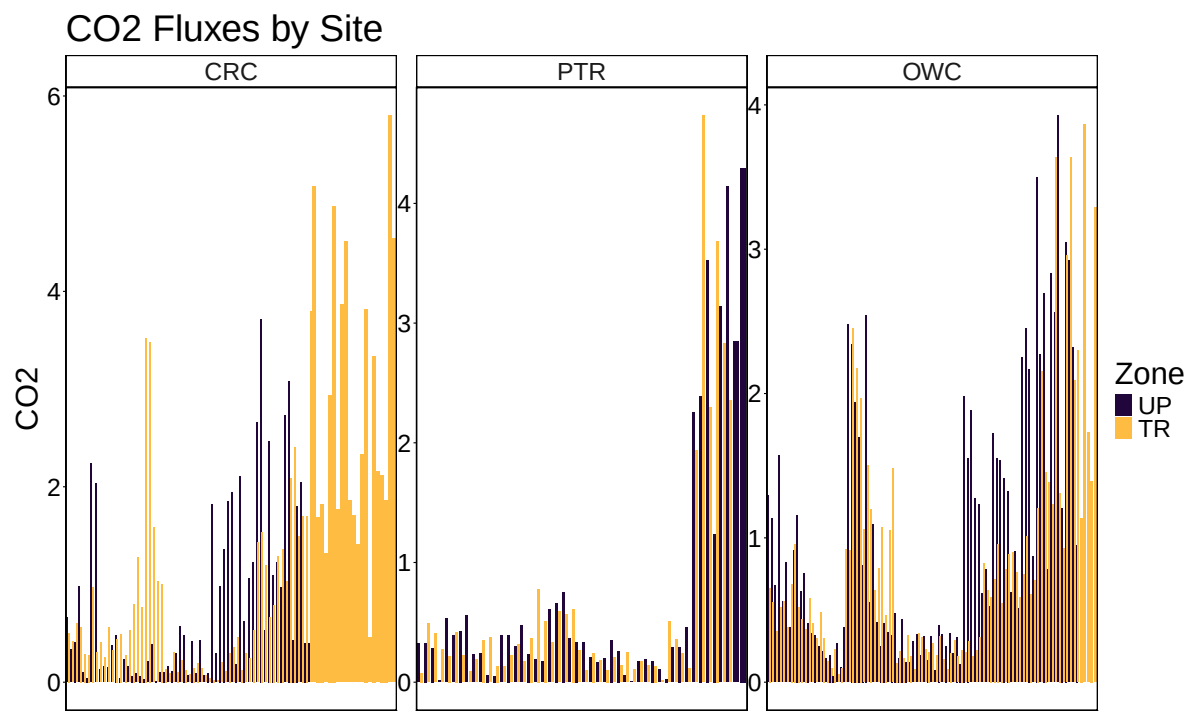
##note that this list has some not synoptic monitoring samples zone-labeled as "Luci" that w

##note that this list has some diurnal variation samples for PTR synoptic veg campaign in 20
```

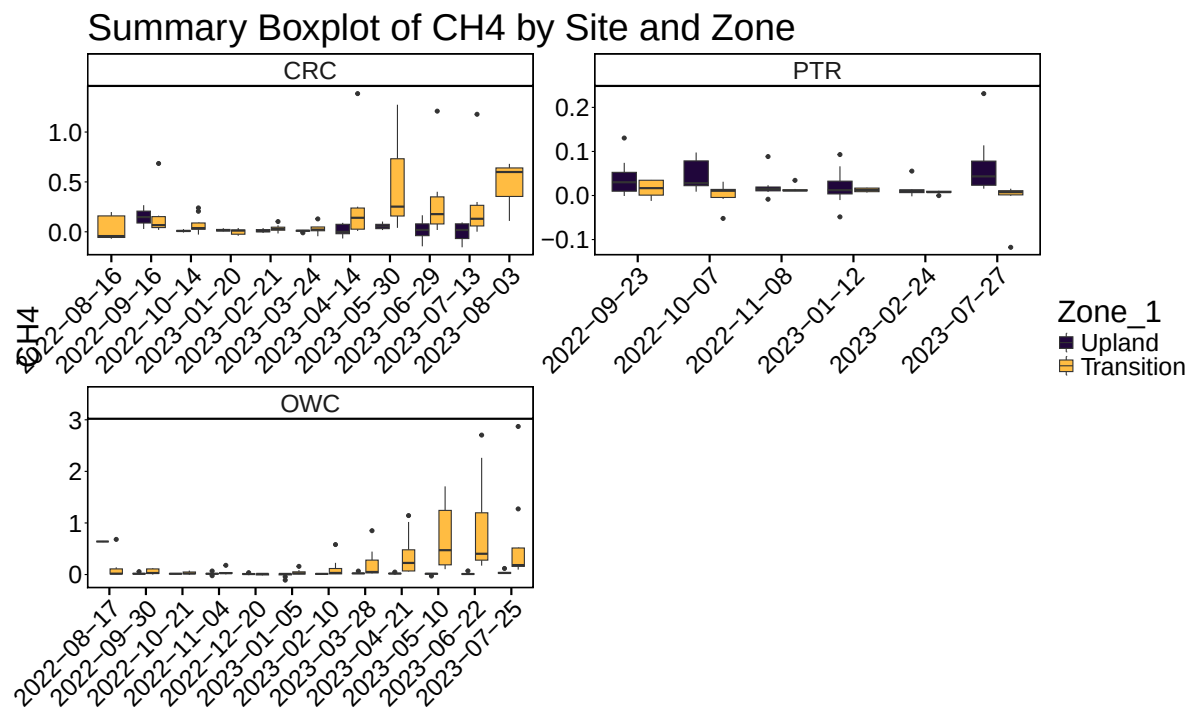
3 Write out files

4 All Data by Year, Site and Zone

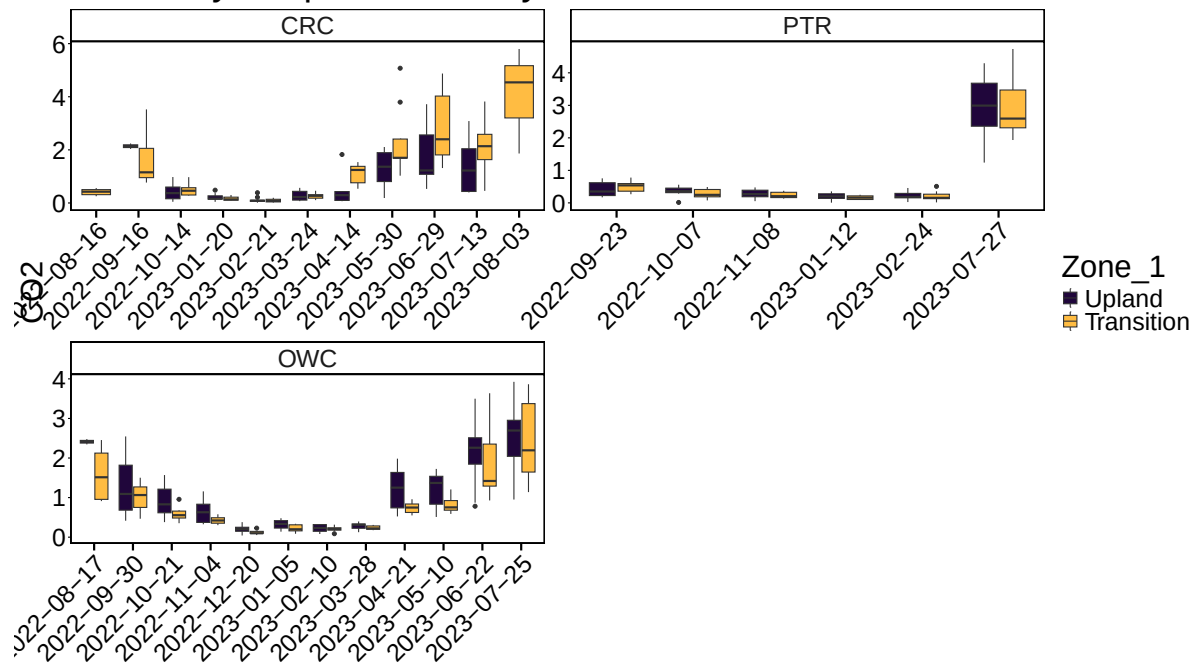




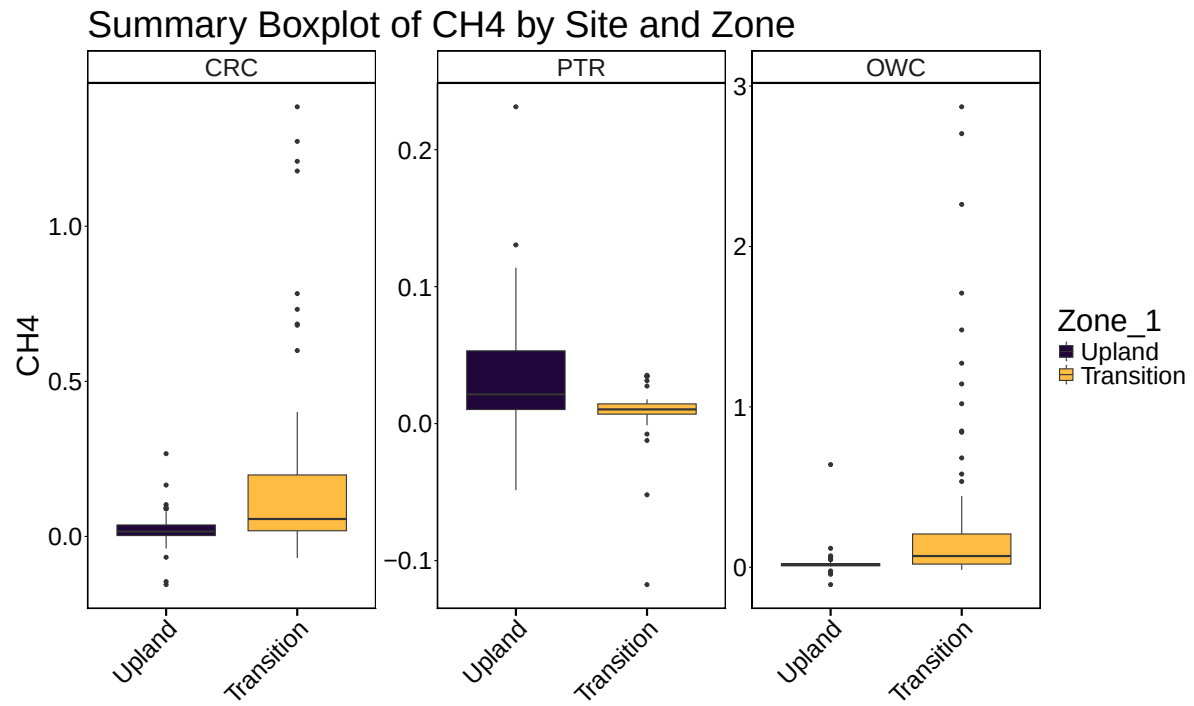
5 Boxplot summary by Date

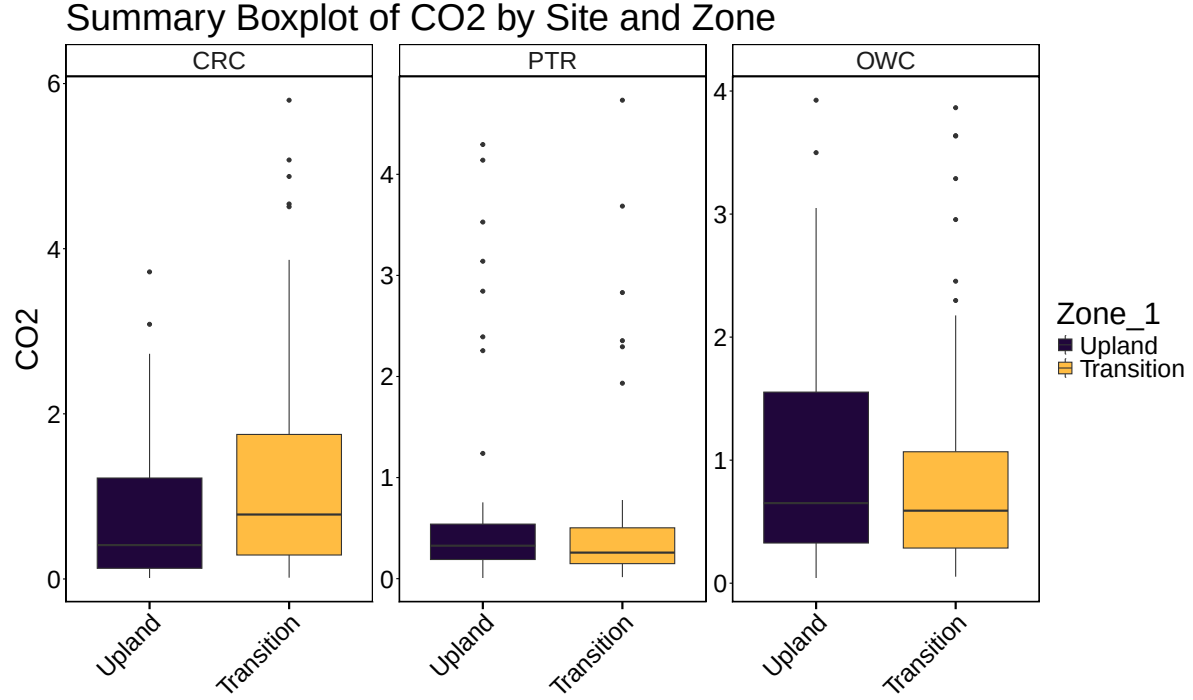


Summary Boxplot of CO2 by Site and Zone



6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 2: Summary Statistics by Site and Zone

Site	Zone	Min CH4	Max CH4	Mean CH4	Min CO2	Max CO2	Mean CO2
CRC	Upland	-0.1552949	0.2668133	0.0222970	0.0116738	3.719250	0.8100245
CRC	Transition	-0.0693972	1.3849897	0.1818432	0.0153453	5.798788	1.3017843
PTR	Upland	-0.0485061	0.2312600	0.0357526	0.0066524	4.294703	0.7386122
PTR	Transition	-0.1175338	0.0352939	0.0074455	0.0152666	4.733226	0.6290828
OWC	Upland	-0.1082825	0.6401738	0.0247291	0.0421910	3.925557	1.0341256
OWC	Transition	-0.0164786	2.8710651	0.2839272	0.0527487	3.864481	0.8639219